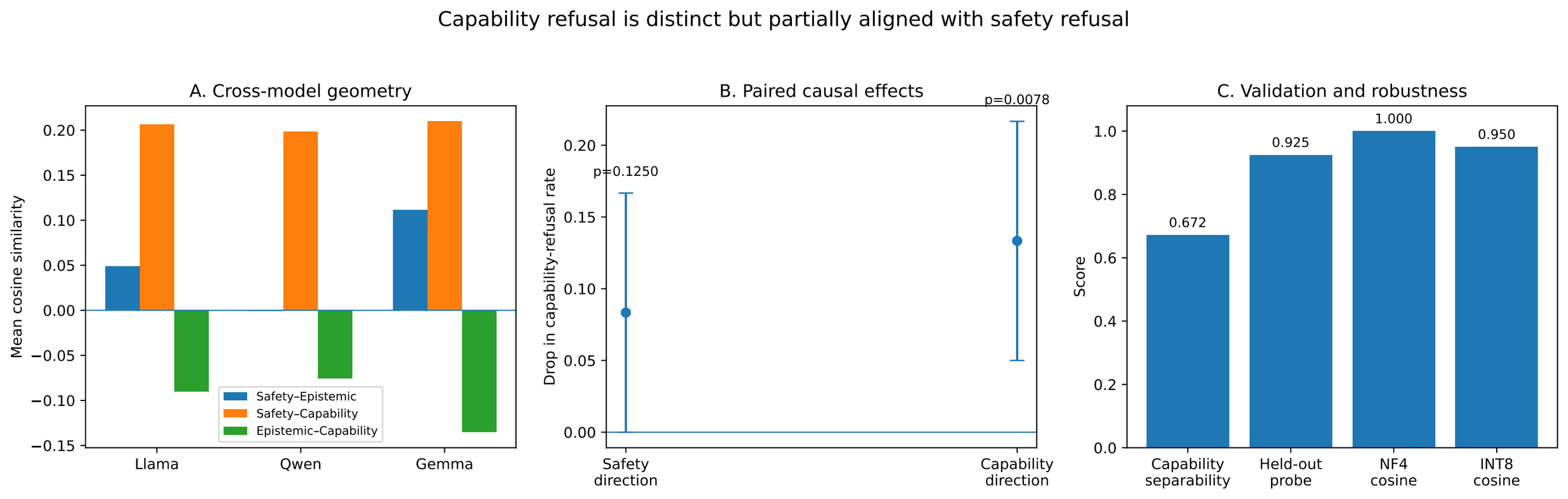
Capability refusal is distinct but partially aligned with safety refusal
A. Cross-model geometry
Mean cosine similarity
0.20
0.15
0.10
0.05
0.00
−0.05
−0.10
−0.15
Llama
Qwen
Gemma
Safety–Epistemic
Safety–Capability
Epistemic–Capability
B. Paired causal effects
Drop in capability-refusal rate
p=0.1250
p=0.0078
0.20
0.15
0.10
0.05
0.00
Safety direction
Capability direction
C. Validation and robustness
Score
1.0
0.8
0.6
0.4
0.2
0.0
0.672
0.925
1.000
0.950
Capability separability
Held-out probe
NF4 cosine
INT8 cosine